Cloze, Frequency, Surprisal, or Plausibility? A Comparative Analysis of Predictors for Local Ambiguity Resolution

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Background: The garden-path (GP) effect is a well-known and frequently studied phenomenon in psycholinguistics, yet the factors influencing its presence and magnitude remain unclear, as the effects often appear inconsistently, only in some studies or for specific structures [1, 2]. Our study investigates three potential predictors of the GP effect in Czech: structural bias, frequency, plausibility, and surprisal. We focus on properties of three sentence segments related to key processing stages: the initial misanalysis of the ambiguous region (AMBMIS), the correct analysis of the ambiguous region (AMBCOR), and the correct analysis of the disambiguating region (DISCOR). To account for potential structural differences, we employed a diverse set of stimuli. We expect that the more frequent, plausible, and predictable AMBMIS is, the more difficult it will be to (re)analyse the GP sentence. The opposite pattern is expected for AMBCOR and DISCOR.
Method: We ran a web-based word-by-word self-paced reading experiment (n=295), collecting reaction times (RTs) and response accuracy (RA) for 66 Czech GP sentences. The items were matched for length but varied in syntactic structure, subjective naturalness, and semantic properties. Each sentence was followed by a yes/no comprehension question targeting the initial misanalysis. Structural bias was estimated using a cloze task (n=115), calculated as the proportion of responses consistent with AMBMIS and AMBCOR; absolute and relative frequency was calculated as the number of occurrences of a subcategorization frame of a specific verb consistent with AMBMIS and AMBCOR using Intercorp v16ud [3]; surprisal estimates of AMBMIS, AMBCOR and DISCOR were calculated using the CzeGPT-2 model; plausibility ratings of AMBMIS, AMBCOR and DISCOR were collected online using a rating scale with a slider (n=111).
Results: We found clear differences in RTs and RA between the GP and non-GP conditions. We ran separate mixed effect models for each predictor in interaction with ambiguity, analysing effects on RA and RTs (see Fig. 1 and 2), and compared the models using their BIC. The cloze score of AMBMIS was the only predictor which showed a significant interaction with ambiguity for both RA ($\beta = -0.344$, $SE = 0.1$, $z = -3.446$, $p < 0.001$) and RTs ($\beta = 0.051$, $SE = 0.012$, $t = 4.384$, $p < 0.001$). There was also a significant interaction between ambiguity and plausibility of AMBCOR for the RA ($\beta = 0.342$, $SE = 0.1$, $z = 3.433$, $p < 0.001$), and of ambiguity with cloze score of AMBCOR ($\beta = -0.053$, $SE = 0.012$, $t = -4.593$, $p < 0.001$) and ambiguity with surprisal of DISCOR ($\beta = 0.044$, $SE = 0.012$, $t = 3.642$, $p < 0.001$) for RTs. Both cloze scores were also consistently assessed as having the best fit to the data according to BIC.
Discussion: Our study yields several important findings. 1) The likelihood of initial misanalysis (i.e. the “structural bias”, measured by cloze scores) is, by far, the best predictor of GP effects in both RTs and RA. 2) Properties of AMBMIS alone are insufficient to explain the variation in GP effect; other parts/stages of analysis–especially AMBCOR–also play a critical role. Yet, many previous studies have overlooked these aspects. 3) Previously reported effects of plausibility or frequency do not replicate when a more diverse stimulus set is used. This is in line with studies showing that different GP structures might be processed differently, and that comprehenders also have individual strategies for dealing with them [1, 4].

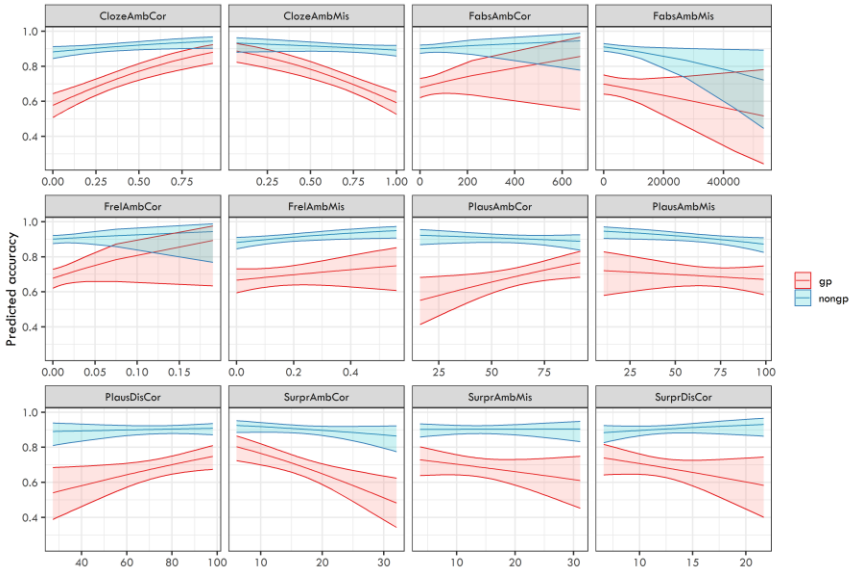

Fig.1: Model predictions for reaction times on the spillover region. Cloze = cloze task, Fabs = absolute frequency, Frel = relative frequency, Plaus = plausibility, Surp = surprisal

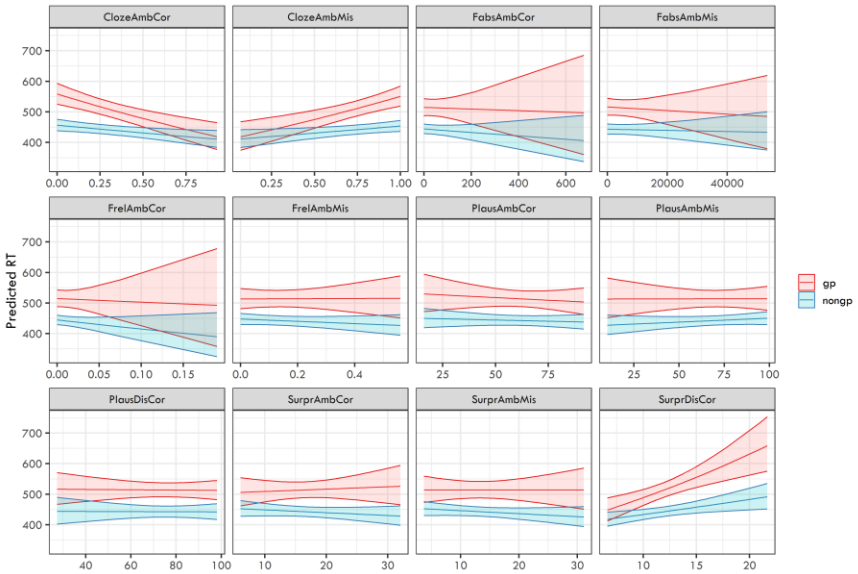

Fig. 2: Model predictions for response accuracy. Cloze = cloze task, Fabs = absolute frequency, Frel = relative frequency, Plaus = plausibility, Surp = surprisal

References

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